

8(a)	any four from: 1 both photosystem I and photosystem II involved ; A both photosystems 2 photoactivation of chlorophyll occurs ; 3 photolysis of water occurs / description ; 4 oxygen production / oxygen-evolving complex involved ; 5 electrons emitted by photosystem I used to reduce NADP or NADP is the final electron acceptor ; 6 photosystem I receives electrons from photosystem II or photosystem II receives electrons from photolysis (of water) ; 7 ETC / electron transport chain, sets up proton gradient or chemiosmosis / ATP produced ;	4
8(b)	GP used to make, amino acids / fatty acids / lipids ; TP used to make, hexose / starch / cellulose / amino acids / glycerol / lipids ;	2
8(c)(i)	$\frac{3.10 - 1.55}{15}$ or $\frac{3.10 - 1.50}{15}$; 0.10 / 0.11 ;	2
8(c)(ii)	any three from: 1 temperature now not limiting or carbon dioxide concentration now limiting ; 2 closure of stomata ; 3 lack of water / AW ; 4 denaturation of enzymes or active site no longer complementary to substrate / no ESCs form ; 5 AVP ; e.g. <i>ref. to optimum temperature</i> / photorespiration occurs	3

Question	Answer	Marks
8(c)(iii)	similar shape but below high intensity curve ;	1

Question	Answer	Marks
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